

An Exact Four-State Theory of Feedback Control with Epistemic Resistance

A pedagogical Stage-I model for sensing, conditional mutual information, susceptibility, and information-response bounds

Working theory note

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Abstract

We develop a deliberately small discrete model for a population subject to feedback control. The population has two coarse variables: a binary social state X_t describing support for the controller target and a binary epistemic state K_t describing whether the population is in a high-information regime. A noisy sensor produces Y_t , a stochastic controller chooses an action U_t , and the next-day state $S_{t+1} = (X_{t+1}, K_{t+1})$ follows an exactly specified Markov kernel. Because the state space contains only four states, all finite-time distributions and information quantities are available without Monte Carlo simulation. The note derives, step by step, the sensing information $I(S_t; Y_t)$, the information actually used by the policy $I(S_t; U_t)$, the action-to-population conditional mutual information $I(U_t; X_{t+1} \mid X_t, K_t)$, the local susceptibility, and a decomposition of conditional action information into an average response term and an epistemic-modulation term. Two rigorous Pinsker-type inequalities then give bounded information-response efficiencies. The purpose is not yet to claim a literal thermodynamic efficiency, but to construct an exact reference system in the spirit of information-thermodynamic feedback bounds and stochastic thermodynamics without an energetic interpretation.

1 Motivation and scope

The long-term objective is to understand resource-constrained feedback control of interacting language-model agents whose decisions depend on both social information and distributed task evidence. Before constructing a mean-field theory for explicit fact spreading, it is useful to have a model small enough that every probability and information quantity can be written down exactly.

The present note is therefore a *Stage-I dummy theory*. It follows the same conceptual ordering as the feedback-information literature: define the stochastic dynamics first, identify what is measured, identify how the measurement changes the policy, and only then ask which information quantity can bound a useful response. Horowitz and Sandberg emphasize that different information measures correspond to different operational measurement scenarios, and hence different numerical “efficiencies” need not have identical physical meaning [1]. Recent work on stochastic thermodynamics of social imitation likewise shows that thermodynamic structure can be exported to social stochastic processes without assigning literal energies, provided the microscopic transition structure is specified carefully [2].

Accordingly, throughout this note we distinguish two claims:

1. **Exact information-response statements.** These follow from probability theory and information inequalities and are rigorous for the model below.
2. **Thermodynamic interpretation.** This requires an explicit reverse process or generalized local detailed balance construction and is postponed to a later stage.

All logarithms are natural unless stated otherwise, so information is measured in *nats*. This convention is important for the constants in Pinsker's inequality.

2 The four-state day/night model

2.1 Population state

At the end of day t the coarse population state is

$$S_t = (X_t, K_t), \quad (1)$$

with

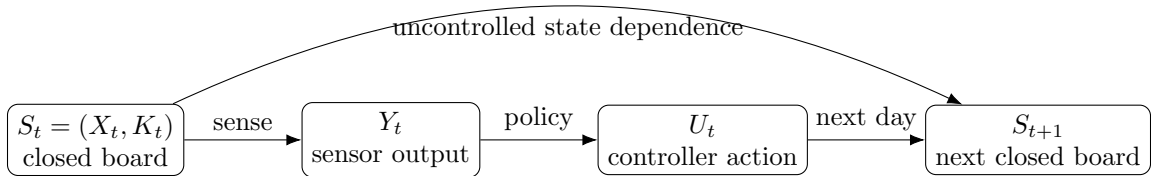
$$X_t \in \{0, 1\}, \quad K_t \in \{0, 1\}. \quad (2)$$

We interpret

- $X_t = 1$: the population supports the controller's target;
- $X_t = 0$: the population opposes the controller's target;
- $K_t = 1$: the population is in a high-epistemic-information state;
- $K_t = 0$: the population is in a low-information state.

For a false-targeting controller, $K = 1$ will be chosen to resist control.

The round clock is



Thus the controller senses after a day closes, chooses an action, and its intervention affects the following day.

2.2 Binary sensing channel

The controller does not see S_t exactly. It receives a binary observation $Y_t \in \{0, 1\}$ through a binary symmetric channel:

$$\mathbb{P}(Y_t = X_t \mid X_t) = 1 - \epsilon, \quad \mathbb{P}(Y_t \neq X_t \mid X_t) = \epsilon, \quad (3)$$

with $0 \leq \epsilon \leq 1/2$. The sensor reads the social state only, not the epistemic state:

$$\mathbb{P}(Y_t = y \mid X_t = x, K_t = k) = \mathbb{P}(Y_t = y \mid X_t = x). \quad (4)$$

This conditional independence will be the key to the sensing-information derivation.

2.3 Stochastic feedback policy

The controller maps the sensor output to a binary action $U_t \in \{0, 1\}$:

$$\mathbb{P}(U_t = 1 \mid Y_t = 0) = \pi_0, \quad \mathbb{P}(U_t = 1 \mid Y_t = 1) = \pi_1. \quad (5)$$

For a controller that intervenes mostly when support for its target is low, one expects $\pi_0 > \pi_1$.

After marginalizing the sensor, the action probability conditional on the true social state is

$$a_0 \equiv \mathbb{P}(U_t = 1 \mid X_t = 0) = (1 - \epsilon)\pi_0 + \epsilon\pi_1, \quad (6)$$

$$a_1 \equiv \mathbb{P}(U_t = 1 \mid X_t = 1) = \epsilon\pi_0 + (1 - \epsilon)\pi_1. \quad (7)$$

Because the sensor ignores K_t , U_t is conditionally independent of K_t given X_t :

$$U_t \perp K_t \mid X_t. \quad (8)$$

2.4 Controlled social dynamics

The next social state is Bernoulli with

$$p_{xku} \equiv \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = u) = \sigma[\theta + J(2x - 1) - gk + hu], \quad (9)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$. The parameters are deliberately interpretable:

- J : persistence of the current social state;
- $g > 0$: epistemic resistance to a false controller;
- $h > 0$: actuation strength;
- θ : baseline bias.

No physical energy is being assigned to the logistic field in Eq. (9); it is simply a convenient stochastic response kernel.

2.5 Coarse epistemic dynamics

The epistemic state follows a two-state acquisition/forgetting process:

$$\mathbb{P}(K_{t+1} = 1 \mid K_t = 0) = \alpha, \quad \mathbb{P}(K_{t+1} = 0 \mid K_t = 1) = \delta. \quad (10)$$

Hence

$$\kappa^* \equiv \mathbb{P}(K = 1)_{\text{stat}} = \frac{\alpha}{\alpha + \delta}. \quad (11)$$

In the later mean-field theory, α and δ will be replaced by effective rates generated by fact transmission and persistence.

2.6 Exact four-by-four transition matrix

Order the four states as

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1). \quad (12)$$

The only potentially confusing notation in the matrix below is the shorthand A_{xk} and B_{xk} . We therefore derive those quantities before writing the matrix.

First average over the controller action

At a fixed current state $(X_t, K_t) = (x, k)$, the next social state still depends on whether the controller acts. Recall

$$p_{xku} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = u). \quad (13)$$

The controller acts with probability

$$a_x = \mathbb{P}(U_t = 1 \mid X_t = x). \quad (14)$$

Therefore, after averaging over the stochastic controller action,

$$\bar{p}_{xk} \equiv \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k) \quad (15)$$

$$= (1 - a_x)p_{xk0} + a_x p_{xk1}. \quad (16)$$

This is simply a two-component mixture: with probability $1 - a_x$ the controller is silent and the transition probability is p_{xk0} ; with probability a_x it acts and the transition probability is p_{xk1} .

We now introduce the abbreviations

$$\boxed{B_{xk} \equiv \bar{p}_{xk}} \quad \text{and} \quad \boxed{A_{xk} \equiv 1 - \bar{p}_{xk}}. \quad (17)$$

Thus they have no additional dynamical meaning:

$$B_{xk} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k), \quad (18)$$

$$A_{xk} = \mathbb{P}(X_{t+1} = 0 \mid X_t = x, K_t = k). \quad (19)$$

The letters A and B are used only to keep the matrix compact.

For example, at current state $(x, k) = (0, 1)$,

$$B_{01} = (1 - a_0)p_{010} + a_0 p_{011}, \quad (20)$$

where p_{010} is the probability of target support tomorrow if the controller is silent, and p_{011} is the corresponding probability if it intervenes. Then

$$A_{01} = 1 - B_{01}. \quad (21)$$

Combine the social and epistemic transitions

The Stage-I model assumes that, conditional on the current state and after averaging over U_t , the next social and epistemic coordinates factorize:

$$\mathbb{P}(X_{t+1} = x', K_{t+1} = k' \mid X_t = x, K_t = k) = \mathbb{P}(X_{t+1} = x' \mid x, k) \mathbb{P}(K_{t+1} = k' \mid k). \quad (22)$$

For instance, starting from $(0, 0)$,

$$\mathbb{P}((0, 0) \rightarrow (0, 0)) = A_{00}(1 - \alpha), \quad (23)$$

$$\mathbb{P}((0, 0) \rightarrow (0, 1)) = A_{00}\alpha, \quad (24)$$

$$\mathbb{P}((0, 0) \rightarrow (1, 0)) = B_{00}(1 - \alpha), \quad (25)$$

$$\mathbb{P}((0, 0) \rightarrow (1, 1)) = B_{00}\alpha. \quad (26)$$

The first factor in each product chooses the next social state; the second factor chooses the next epistemic state.

With this notation the exact transition matrix is

$$M = \begin{pmatrix} A_{00}(1-\alpha) & A_{00}\alpha & B_{00}(1-\alpha) & B_{00}\alpha \\ A_{01}\delta & A_{01}(1-\delta) & B_{01}\delta & B_{01}(1-\delta) \\ A_{10}(1-\alpha) & A_{10}\alpha & B_{10}(1-\alpha) & B_{10}\alpha \\ A_{11}\delta & A_{11}(1-\delta) & B_{11}\delta & B_{11}(1-\delta) \end{pmatrix}. \quad (27)$$

If one prefers to avoid the A/B shorthand completely, the first row is equivalently

$$[(1-\bar{p}_{00})(1-\alpha), (1-\bar{p}_{00})\alpha, \bar{p}_{00}(1-\alpha), \bar{p}_{00}\alpha], \quad (28)$$

and the remaining rows follow in exactly the same way.

Finally, if

$$\boldsymbol{\mu}_t = (\mathbb{P}_t(0,0), \mathbb{P}_t(0,1), \mathbb{P}_t(1,0), \mathbb{P}_t(1,1)), \quad (29)$$

then

$$\boxed{\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t M.} \quad (30)$$

Thus all finite-time observables are obtained exactly by matrix multiplication.

3 Exact finite-time and stationary state distribution

Before introducing sensing information, it is useful to remove a possible source of arbitrariness. In the sensing derivation below we write

$$q_t = \mathbb{P}(X_t = 1). \quad (31)$$

This quantity does *not* need to be chosen by hand once the four-state model has been specified. It is determined exactly by the Markov dynamics.

3.1 Finite-time state distribution

Recall that the full four-state distribution is

$$\boldsymbol{\mu}_t = (\mu_{00}(t), \mu_{01}(t), \mu_{10}(t), \mu_{11}(t)), \quad (32)$$

where

$$\mu_{xk}(t) = \mathbb{P}(X_t = x, K_t = k). \quad (33)$$

The exact time evolution is

$$\boxed{\boldsymbol{\mu}_t = \boldsymbol{\mu}_0 M^t.} \quad (34)$$

Therefore the social marginal is obtained directly from the two states having $X_t = 1$:

$$\boxed{q_t = \mathbb{P}(X_t = 1) = \mu_{10}(t) + \mu_{11}(t).} \quad (35)$$

Equivalently,

$$q_t = \boldsymbol{\mu}_0 M^t \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}. \quad (36)$$

Likewise, the epistemic marginal is

$$\boxed{\kappa_t = \mathbb{P}(K_t = 1) = \mu_{01}(t) + \mu_{11}(t).} \quad (37)$$

Thus both coordinates that later enter the information analysis are endogenous observables of the same exact four-state chain.

3.2 Stationary distribution

If the chain is ergodic, the distribution converges to a unique stationary distribution

$$\boxed{\boldsymbol{\mu}^* = \boldsymbol{\mu}^* M, \quad \sum_{x,k} \mu_{xk}^* = 1.} \quad (38)$$

In matrix language, $\boldsymbol{\mu}^*$ is the normalized left eigenvector of M with eigenvalue one. Once it is known, the stationary social and epistemic marginals follow immediately:

$$\boxed{q^* = \mu_{10}^* + \mu_{11}^*}, \quad (39)$$

$$\boxed{\kappa^* = \mu_{01}^* + \mu_{11}^*}. \quad (40)$$

The epistemic marginal has an especially simple closed form because K_t evolves autonomously according to the two-state acquisition/forgetting chain. Stationarity requires

$$(1 - \kappa^*)\alpha = \kappa^*\delta, \quad (41)$$

and therefore

$$\boxed{\kappa^* = \frac{\alpha}{\alpha + \delta}}. \quad (42)$$

The social marginal q^* is generally coupled to the full four-state stationary distribution because the transition probability of X depends on both X and K .

3.3 Concrete stationary solution for the reference parameters

For the reference parameter set used in the plots,

$$\epsilon = 0.2, \quad \pi_0 = 0.85, \quad \pi_1 = 0.15, \quad \theta = -0.2, \quad J = 1, \quad g = 1.5, \quad h = 1.4, \quad \alpha = 0.25, \quad \delta = 0.12, \quad (43)$$

the exact transition matrix is

$$M \simeq \begin{pmatrix} 0.4069 & 0.1356 & 0.3431 & 0.1144 \\ 0.0996 & 0.7301 & 0.0204 & 0.1499 \\ 0.1868 & 0.0623 & 0.5632 & 0.1877 \\ 0.0685 & 0.5022 & 0.0515 & 0.3778 \end{pmatrix}. \quad (44)$$

Solving Eq. (38) gives

$$\boxed{\boldsymbol{\mu}^* \simeq (0.1561, 0.4805, 0.1682, 0.1952)}. \quad (45)$$

Hence

$$q^* = 0.1682 + 0.1952 \simeq \boxed{0.3634}, \quad (46)$$

$$\kappa^* = 0.4805 + 0.1952 \simeq \boxed{0.6757}, \quad (47)$$

and indeed

$$\frac{\alpha}{\alpha + \delta} = \frac{0.25}{0.25 + 0.12} \simeq 0.6757. \quad (48)$$

Figure 1 visualizes this stationary occupancy. It is important to distinguish the four-state distribution $\boldsymbol{\mu}^*$ from the scalar marginals q^* and κ^* : the latter are obtained only after summing over the appropriate components of the former.

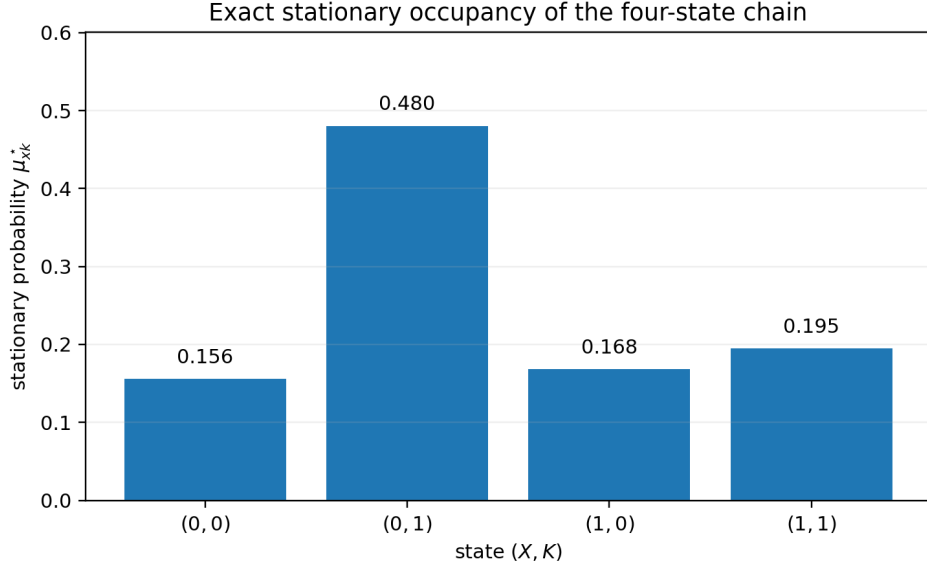


Figure 1: Exact stationary distribution of the four-state chain for the reference parameter set. The stationary social support is $q^* = \mu_{10}^* + \mu_{11}^* \simeq 0.3634$, while the stationary high-information share is $\kappa^* = \mu_{01}^* + \mu_{11}^* \simeq 0.6757$.

3.4 Why this matters for the information quantities

The stationary distribution removes the need to choose an illustrative value of q when we want a model-specific prediction. For example, the stationary sensing information is

$$I_{\text{sens}}^* = h[\epsilon + (1 - 2\epsilon)q^*] - h(\epsilon), \quad (49)$$

where q^* is obtained from the same transition matrix M . Likewise, the stationary action-to-population conditional mutual information is

$$T_{\pi}^* = \sum_{x,k} \mu_{xk}^* T_{xk}. \quad (50)$$

Thus there are two complementary viewpoints throughout the note:

- **local quantities**, such as T_{xk} or χ_{xk} , which describe the response from one specified state;
- **population-level quantities**, which average those local responses with the exact finite-time distribution μ_t or the stationary distribution μ^* .

The next section derives the sensing information for a generic current q_t ; in the full four-state model, that q_t is supplied exactly by Eq. (35), and at stationarity by q^* .

4 Sensing mutual information, derived step by step

This section derives the formula that is easy to quote but easy to lose intuition for:

$$I_{\text{sens}}(t) = I(S_t; Y_t). \quad (51)$$

4.1 Step 1: start from the definition

For arbitrary discrete variables A and B ,

$$I(A; B) = \sum_{a,b} p(a, b) \ln \frac{p(a, b)}{p(a)p(b)} = \sum_{a,b} p(a, b) \ln \frac{p(b | a)}{p(b)}. \quad (52)$$

Rearranging the logarithm gives the entropy identity

$$I(A; B) = H(B) - H(B | A). \quad (53)$$

For the sensor,

$$I(S_t; Y_t) = H(Y_t) - H(Y_t | S_t). \quad (54)$$

This already gives the interpretation: sensing information is the uncertainty in the sensor output minus the uncertainty that remains once the true state is known.

4.2 Step 2: why the epistemic variable disappears from the sensor term

Because $S_t = (X_t, K_t)$,

$$I(S_t; Y_t) = I(X_t, K_t; Y_t). \quad (55)$$

The chain rule for mutual information gives

$$I(X_t, K_t; Y_t) = I(X_t; Y_t) + I(K_t; Y_t | X_t). \quad (56)$$

But Eq. (4) says that once X_t is known, the sensor output carries no further information about K_t . Therefore

$$I(K_t; Y_t | X_t) = 0, \quad (57)$$

and hence

$$\boxed{I(S_t; Y_t) = I(X_t; Y_t)}. \quad (58)$$

This equality is not a generic property of epistemic feedback systems. It follows specifically from our modeling choice that the controller senses only the social state.

4.3 Step 3: compute the sensor-output distribution

Let

$$q_t \equiv \mathbb{P}(X_t = 1). \quad (59)$$

The probability that the sensor reports $Y_t = 1$ is obtained by the law of total probability:

$$r_t \equiv \mathbb{P}(Y_t = 1) \quad (60)$$

$$= \mathbb{P}(Y_t = 1 | X_t = 1)\mathbb{P}(X_t = 1) + \mathbb{P}(Y_t = 1 | X_t = 0)\mathbb{P}(X_t = 0) \quad (61)$$

$$= (1 - \epsilon)q_t + \epsilon(1 - q_t) \quad (62)$$

$$= \boxed{\epsilon + (1 - 2\epsilon)q_t}. \quad (63)$$

Thus Y_t is Bernoulli with parameter r_t .

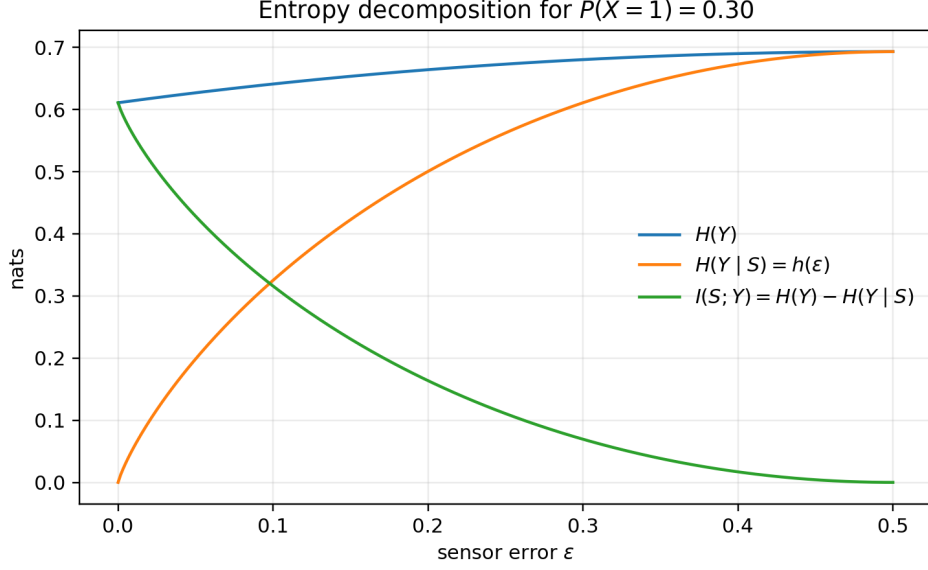


Figure 2: The sensing information is literally the vertical difference between the output entropy $H(Y)$ and the residual sensor noise $H(Y | S)$. The example uses $P(X = 1) = 0.30$.

4.4 Step 4: compute the two entropies

Define the binary entropy in nats

$$h(p) = -p \ln p - (1 - p) \ln(1 - p). \quad (64)$$

Since $Y_t \sim \text{Bernoulli}(r_t)$,

$$H(Y_t) = h(r_t). \quad (65)$$

Conditional on either value of X_t , the sensor flips the correct value with probability ϵ . Therefore

$$H(Y_t | X_t = x) = h(\epsilon) \quad \text{for both } x = 0, 1. \quad (66)$$

Averaging over X_t changes nothing:

$$H(Y_t | S_t) = H(Y_t | X_t) = h(\epsilon). \quad (67)$$

4.5 Step 5: closed form

Substituting into Eq. (54),

$$\boxed{I_{\text{sens}}(t) = I(S_t; Y_t) = h(\epsilon + (1 - 2\epsilon)q_t) - h(\epsilon).} \quad (68)$$

This is exact for any distribution over the four states; only the marginal $q_t = P(X_t = 1)$ is needed.

Two checks are useful:

- If $\epsilon = 1/2$, then Y_t is independent random noise and $I_{\text{sens}} = 0$.
- If $\epsilon = 0$, then $Y_t = X_t$ exactly and $I_{\text{sens}} = H(X_t) = h(q_t)$.

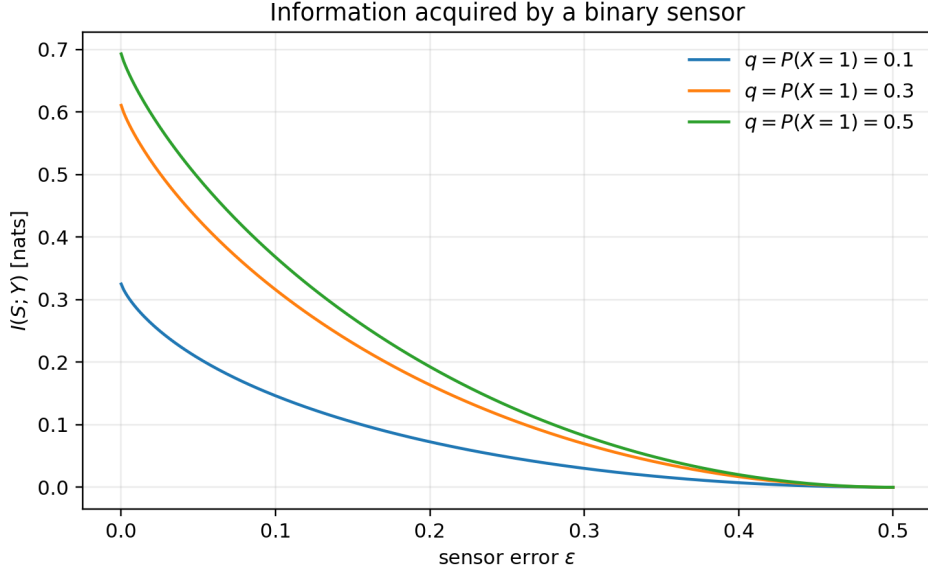


Figure 3: Exact sensing mutual information from Eq. (68). A balanced social state carries more potential information than a nearly deterministic social state, and all sensing information vanishes at $\epsilon = 1/2$.

5 How much of the sensed information is actually used?

Sensing does not imply that the controller uses all available information. The effective state-dependent action probabilities are a_0, a_1 from Eq. (7). With $q_t = P(X_t = 1)$, the marginal intervention probability is

$$\bar{a}_t = (1 - q_t)a_0 + q_t a_1. \quad (69)$$

Since U_t is Bernoulli,

$$H(U_t) = h(\bar{a}_t). \quad (70)$$

Conditioned on the social state,

$$H(U_t | X_t) = (1 - q_t)h(a_0) + q_t h(a_1). \quad (71)$$

Because $U_t \perp K_t | X_t$,

$$I(S_t; U_t) = I(X_t; U_t) = h(\bar{a}_t) - (1 - q_t)h(a_0) - q_t h(a_1). \quad (72)$$

The Markov chain $S_t \rightarrow Y_t \rightarrow U_t$ implies by the data-processing inequality

$$I(S_t; U_t) \leq I(S_t; Y_t). \quad (73)$$

Thus $I(S; Y)$ is acquired state information, whereas $I(S; U)$ is the portion that survives the policy map and becomes state-dependent action.

Figure 4 makes this distinction concrete. Parameterize the policy by a contrast $d \in [0, 1]$,

$$\pi_0 = \frac{1 + d}{2}, \quad \pi_1 = \frac{1 - d}{2}. \quad (74)$$

At $d = 0$ the controller ignores the sensor and acts with probability $1/2$, so $I(S; U) = 0$. At $d = 1$ the action is a deterministic relabeling of the sensor output, so no additional information is lost and $I(S; U) = I(S; Y)$.

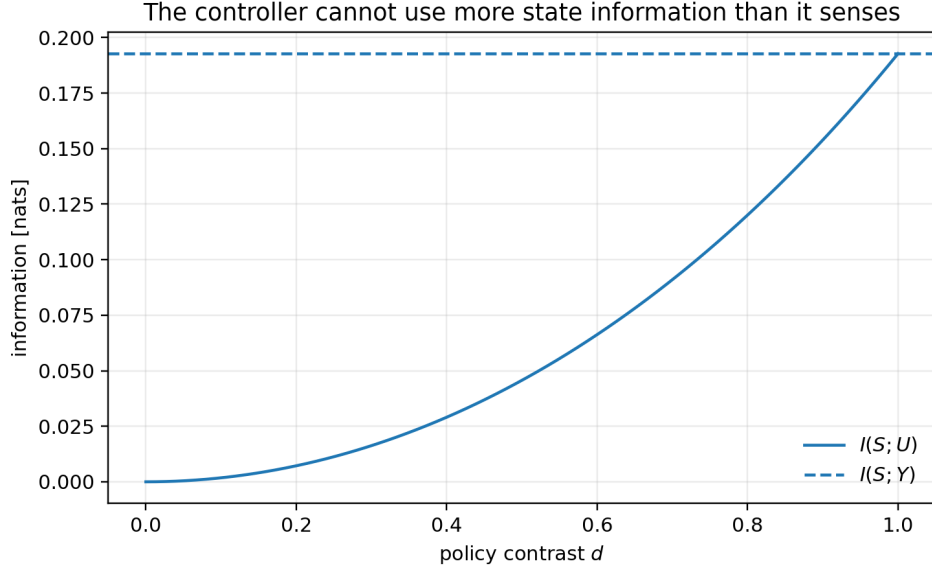


Figure 4: Information available to the controller versus information expressed in its action for $\epsilon = 0.2$ and $P(X = 1) = 0.5$. The policy cannot create state information that was absent from the sensor.

6 Action-to-population conditional mutual information

The sensing information is about the direction $S_t \rightarrow Y_t$. The quantity used in the current control analysis asks the opposite operational question: once the current state is fixed, how much does the controller action tell us about the next population state?

We define

$$T_{\pi}^{X|K}(t) \equiv I(U_t; X_{t+1} \mid X_t, K_t) = I(U_t; X_{t+1} \mid S_t). \quad (75)$$

6.1 Exact local expression: what exactly is T_{xk} ?

Fix a current state

$$S_t = s = (x, k). \quad (76)$$

At this point x and k are not random anymore: we are asking what the controller can do *from this particular social and epistemic state*.

For compactness write

$$a \equiv a_x = \mathbb{P}(U_t = 1 \mid X_t = x), \quad (77)$$

and define the two next-state probabilities

$$p_0 \equiv p_{xk0} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = 0), \quad (78)$$

$$p_1 \equiv p_{xk1} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = 1). \quad (79)$$

Thus p_0 is the next-day target-support probability if the controller is silent, while p_1 is the corresponding probability if it acts.

Marginalizing over the stochastic controller action gives

$$\bar{p} = (1 - a)p_0 + ap_1 = \mathbb{P}(X_{t+1} = 1 \mid S_t = s). \quad (80)$$

The local action-to-next-state conditional mutual information is

$$T_s \equiv I(U_t; X_{t+1} \mid S_t = s) \quad (81)$$

$$= H(X_{t+1} \mid S_t = s) - H(X_{t+1} \mid U_t, S_t = s) \quad (82)$$

$$= \boxed{h(\bar{p}) - (1 - a)h(p_0) - ah(p_1)}. \quad (83)$$

Because every state s is one of the pairs (x, k) , from this point onward we use the equivalent notation

$$\boxed{T_{xk} \equiv T_s \quad \text{for } s = (x, k)}. \quad (84)$$

Thus T_{xk} is *not* another dynamical variable. It is simply the local CMI evaluated when the current social state is x and the current epistemic state is k :

$$\boxed{T_{xk} = I(U_t; X_{t+1} \mid X_t = x, K_t = k)}. \quad (85)$$

For the logistic transition kernel, its parameter dependence is completely explicit. Define

$$z_{xk} = \theta + J(2x - 1) - gk. \quad (86)$$

Then

$$p_0 = \sigma(z_{xk}), \quad p_1 = \sigma(z_{xk} + h), \quad (87)$$

while the action probability is

$$a_x = \begin{cases} (1 - \epsilon)\pi_0 + \epsilon\pi_1, & x = 0, \\ \epsilon\pi_0 + (1 - \epsilon)\pi_1, & x = 1. \end{cases} \quad (88)$$

Consequently,

$$\boxed{T_{xk} = h((1 - a_x)\sigma(z_{xk}) + a_x\sigma(z_{xk} + h)) - (1 - a_x)h(\sigma(z_{xk})) - a_xh(\sigma(z_{xk} + h))}. \quad (89)$$

Here the italic $h(\cdot)$ is the binary entropy function introduced earlier, whereas the parameter h inside $\sigma(z_{xk} + h)$ is the controller actuation strength. The shared symbol is conventional but visually unfortunate; the distinction is always clear from whether h is used as a function or as an additive parameter.

Equation (89) makes the roles of the three most important parameters transparent:

- h changes how far apart the action and no-action next-state distributions are;
- g changes the baseline field only when $k = 1$, and therefore controls epistemic resistance;
- ϵ does not enter the social transition kernel directly, but changes a_x through the sensing-policy channel and therefore changes the amount of action support available at a fixed state.

Figure 5 shows T_{00} and T_{01} as the actuation strength varies. When $h = 0$, action and silence generate the same next-state distribution, so both local CMIs are exactly zero. As h increases, the two action-conditioned distributions separate and T_{xk} increases. The high-information state $K = 1$ is shifted by the epistemic-resistance term $-g$, so its information curve can be displaced relative to the low-information state.

Figure 6 uses the same model rather than arbitrary Bernoulli probabilities and compares the exact T_{01} with the Pinsker response term derived later. This is useful because it shows that the conditional mutual information is the full distributional quantity, while the susceptibility term captures only the coherent shift in the mean.

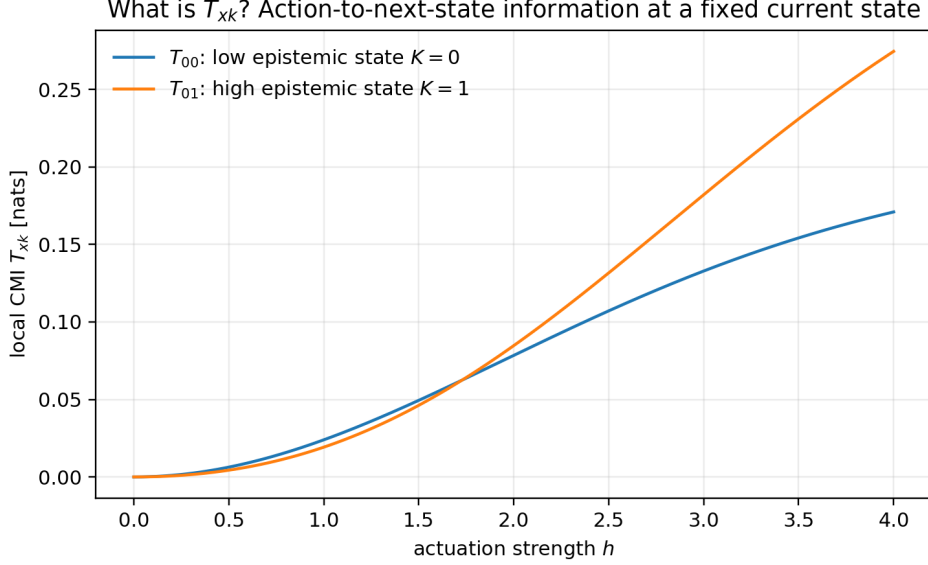


Figure 5: Local conditional mutual information as a function of actuation strength for the reference model, fixing $x = 0$, $\epsilon = 0.2$, and $g = 1.5$. T_{00} and T_{01} mean exactly $I(U_t; X_{t+1} \mid X_t = 0, K_t = 0)$ and $I(U_t; X_{t+1} \mid X_t = 0, K_t = 1)$.

The effect of epistemic resistance is shown directly in Figure 7. Increasing g shifts the uncontrolled decision field for $K = 1$. Whether this suppresses or enhances local CMI depends on the actuation strength: the information is largest when action and silence produce distinguishable but non-saturated next-state distributions.

Figure 8 summarizes the same effect over the (h, g) plane. This is the finite-state analogue of the response surfaces we ultimately want in the mean-field theory: the control channel is strongest in a band where controller strength is sufficient to compete with epistemic resistance without trivially saturating the next-state decision.

Finally, sensor error ϵ plays a conceptually different role. Figure 9 compares sensing information, policy information, and action-to-state conditional information while holding the current state distribution fixed. As $\epsilon \rightarrow 1/2$, the sensor becomes useless and $I(S; Y)$ and $I(S; U)$ vanish. However, $I(U; X_{t+1} \mid X_t, K_t)$ need not vanish: a controller can still randomly act, and if action changes the next-state distribution then the action remains informative about the outcome. This is why sensing MI and transfer-style CMI should not be interpreted as the same resource.

The state-averaged quantity is now simply the occupancy-weighted average of these local channels:

$$T_{\pi}^{X|K}(t) = \sum_{x,k} \mu_t(x, k) T_{xk}. \quad (90)$$

In words, $\mu_t(x, k)$ tells us how often the system is currently in state (x, k) , and T_{xk} tells us how informative the controller action is about the next social state *when the system is in that state*. Equation (90) averages those local quantities according to how frequently the states are occupied.

No estimator is needed in this four-state model. Equation (90) is the exact population value against which a finite-sample estimator can later be tested.

6.2 Why condition on the epistemic state?

One may instead calculate

$$T_{\pi}^X(t) = I(U_t; X_{t+1} \mid X_t), \quad (91)$$

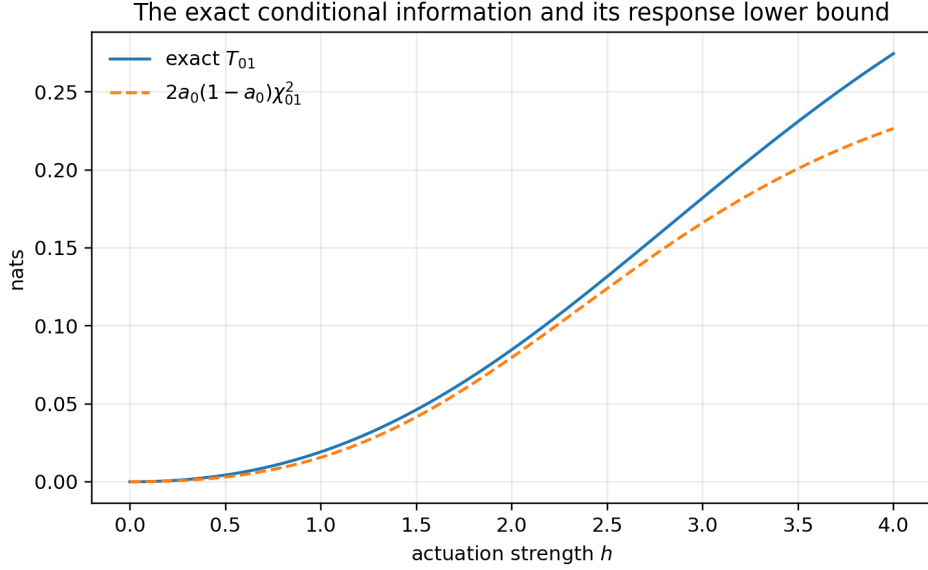


Figure 6: For the high-information state $(x, k) = (0, 1)$, the exact local CMI is compared with the response lower bound $2a_0(1 - a_0)\chi_{01}^2$. Both are zero when $h = 0$ because the controller action then has no effect on the next-state distribution.

which mixes high- and low-information populations having the same social state. The difference between the two quantities follows exactly from the chain rule. On one hand,

$$I(U; X', K | X) = I(U; K | X) + I(U; X' | X, K). \quad (92)$$

Equation (8) sets the first term to zero, so this equals $I(U; X' | X, K)$. On the other hand,

$$I(U; X', K | X) = I(U; X' | X) + I(U; K | X, X'). \quad (93)$$

Equating the two expansions gives

$$\boxed{I(U; X' | X, K) = I(U; X' | X) + I(U; K | X, X') \geq I(U; X' | X).} \quad (94)$$

The extra term is not a nuisance: it quantifies precisely how much controller-action information is hidden when populations with different epistemic states are pooled together. This is the theoretical reason to condition empirical CMI on a coarse epistemic coordinate.

7 Susceptibility: the coherent response to intervention

At fixed state $s = (x, k)$ define the local susceptibility

$$\chi_{xk} \equiv \mathbb{E}[X_{t+1} | x, k, U_t = 1] - \mathbb{E}[X_{t+1} | x, k, U_t = 0] \quad (95)$$

$$= \boxed{p_{xk1} - p_{xk0}}. \quad (96)$$

For the logistic kernel, let

$$z_{xk} = \theta + J(2x - 1) - gk. \quad (97)$$

Then

$$\boxed{\chi_{xk} = \sigma(z_{xk} + h) - \sigma(z_{xk})}. \quad (98)$$

For weak actuation h ,

$$\chi_{xk} = h\sigma(z_{xk})[1 - \sigma(z_{xk})] + \mathcal{O}(h^2). \quad (99)$$

Hence response is small when the next decision is already almost deterministic and largest in an intermediate decision region. For finite h , the maximum is centered near $z = -h/2$.

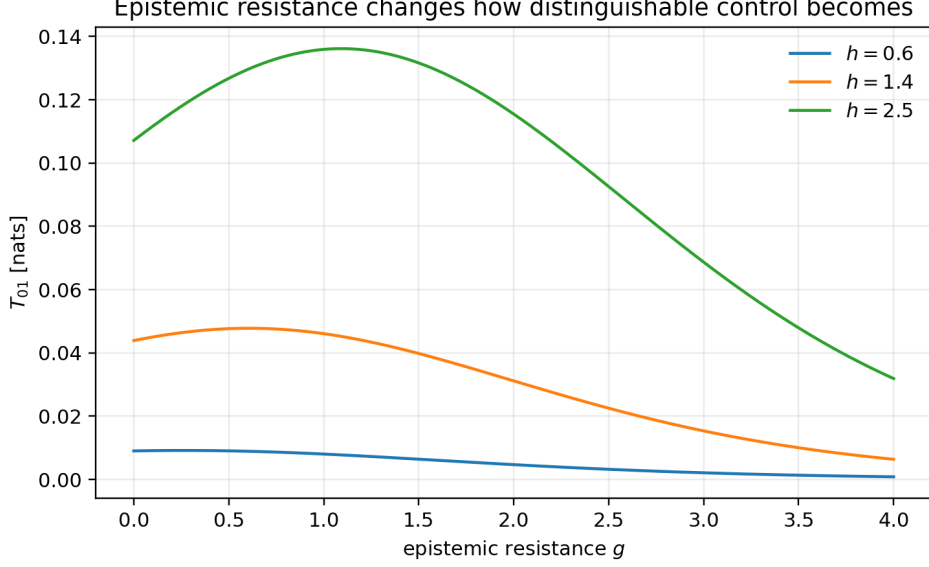


Figure 7: Local CMI T_{01} versus epistemic resistance g for several controller strengths h , with $x = 0$ and $\epsilon = 0.2$. Epistemic resistance does not simply “reduce information” monotonically; instead it moves the system through or away from the decision-sensitive region.

8 A rigorous information-response bound

The conditional mutual information in Eq. (83) measures the full distributional distinction between action-conditioned next states. Susceptibility measures only their coherent difference in mean target support. Pinsker’s inequality relates the two.

8.1 Step 1: write local mutual information as a KL mixture

At a fixed state s , let P_0 and P_1 denote the distributions of X_{t+1} under $U = 0$ and $U = 1$, and let

$$\bar{P} = (1 - a)P_0 + aP_1. \quad (100)$$

Then

$$I(U; X' | s) = (1 - a)D_{\text{KL}}(P_0 \| \bar{P}) + aD_{\text{KL}}(P_1 \| \bar{P}). \quad (101)$$

8.2 Step 2: apply Pinsker’s inequality

In nats,

$$D_{\text{KL}}(P \| Q) \geq 2\text{TV}(P, Q)^2. \quad (102)$$

Because $\bar{P} = (1 - a)P_0 + aP_1$,

$$\text{TV}(P_0, \bar{P}) = a \text{TV}(P_0, P_1), \quad (103)$$

$$\text{TV}(P_1, \bar{P}) = (1 - a) \text{TV}(P_0, P_1). \quad (104)$$

Substituting these into Eq. (101) yields

$$I(U; X' | s) \geq 2a(1 - a)\text{TV}(P_0, P_1)^2. \quad (105)$$

For a binary next state, total variation is exactly the absolute difference of the Bernoulli probabilities:

$$\text{TV}(P_0, P_1) = |p_1 - p_0| = |\chi_s|. \quad (106)$$

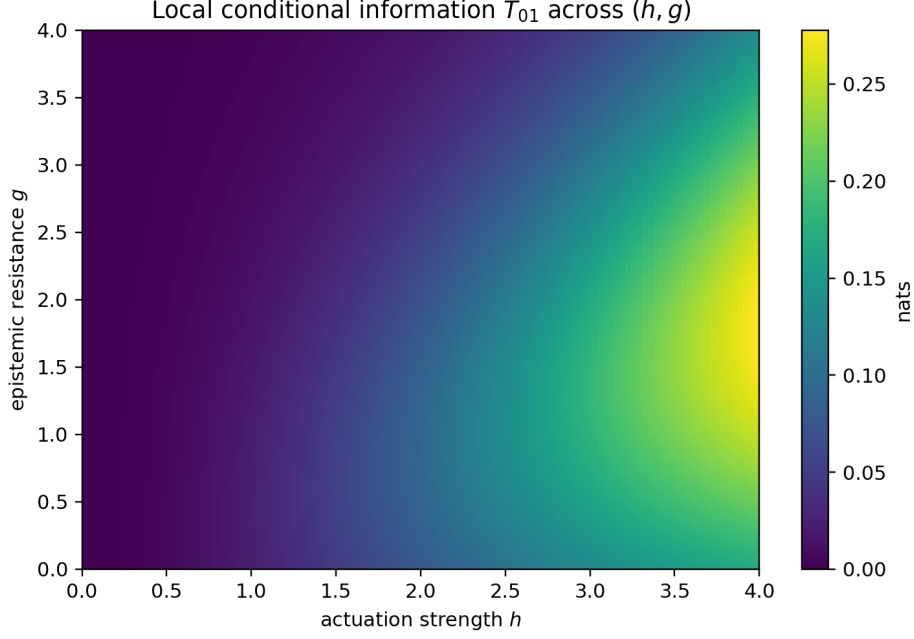


Figure 8: Local action-to-next-state information T_{01} across actuation strength h and epistemic resistance g for fixed $x = 0$ and $\epsilon = 0.2$. The high-information state is easiest to distinguish under action versus silence in an intermediate competition region.

Therefore

$$I(U; X' | S = s) \geq 2a_s(1 - a_s)\chi_s^2. \quad (107)$$

8.3 Step 3: average over states

Averaging Eq. (107) over S_t gives

$$T_\pi^{X|K}(t) \geq 2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2. \quad (108)$$

This motivates the bounded information-response efficiency

$$\eta_{\text{IR}}(t) \equiv \frac{2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2}{I(U_t; X_{t+1} | S_t)} \leq 1. \quad (109)$$

It asks: *what fraction of the action-dependent distributional information is expressed as a coherent mean shift in the chosen population observable?*

Bits versus nats. Empirical analyses often report MI in bits. Since $I_{\text{nats}} = (\ln 2) I_{\text{bits}}$, Eq. (107) becomes

$$I_{\text{bits}}(U; X' | s) \geq \frac{2}{\ln 2} a(1 - a) \chi_s^2. \quad (110)$$

The conversion factor must be included before comparing a bit-valued empirical CMI with the Pinsker numerator.



Figure 9: The effect of sensor error ϵ for a fixed current state distribution. Sensing information $I(S; Y)$ and policy information $I(S; U)$ go to zero when $\epsilon = 1/2$. The action-to-state quantity $I(U; X_{t+1} | X_t, K_t)$ can remain nonzero because it measures whether different actions produce distinguishable next-state distributions, not whether the action was chosen using useful sensing.

9 How epistemic state modulates controllability

The previous section bound the controller’s mean effect at each full state $s = (x, k)$. A separate and scientifically central question is whether *the same controller action has different consequences depending on what the population knows*. This is the point of conditioning on the epistemic state in

$$I(U_t; X_{t+1} | X_t, K_t). \quad (111)$$

Here we make that statement quantitative.

9.1 Step 1: local susceptibilities at fixed social state

Fix the current social state $X_t = x$. There are then two relevant local susceptibilities,

$$\chi_{x0} = \mathbb{E}[X_{t+1} | X_t = x, K_t = 0, U_t = 1] - \mathbb{E}[X_{t+1} | X_t = x, K_t = 0, U_t = 0], \quad (112)$$

and

$$\chi_{x1} = \mathbb{E}[X_{t+1} | X_t = x, K_t = 1, U_t = 1] - \mathbb{E}[X_{t+1} | X_t = x, K_t = 1, U_t = 0]. \quad (113)$$

If these are different, the controller’s ability to move the population depends on epistemic state. Figure 12 shows the basic phenomenon in the logistic model: as the epistemic-resistance parameter g increases, the high-information state $K = 1$ becomes increasingly resistant, and the response gap

$$\Delta\chi_x \equiv \chi_{x1} - \chi_{x0} \quad (114)$$

grows in magnitude.

9.2 Step 2: average over the epistemic state at fixed social state

Let

$$r_x \equiv \mathbb{P}(K_t = 1 | X_t = x) \quad (115)$$

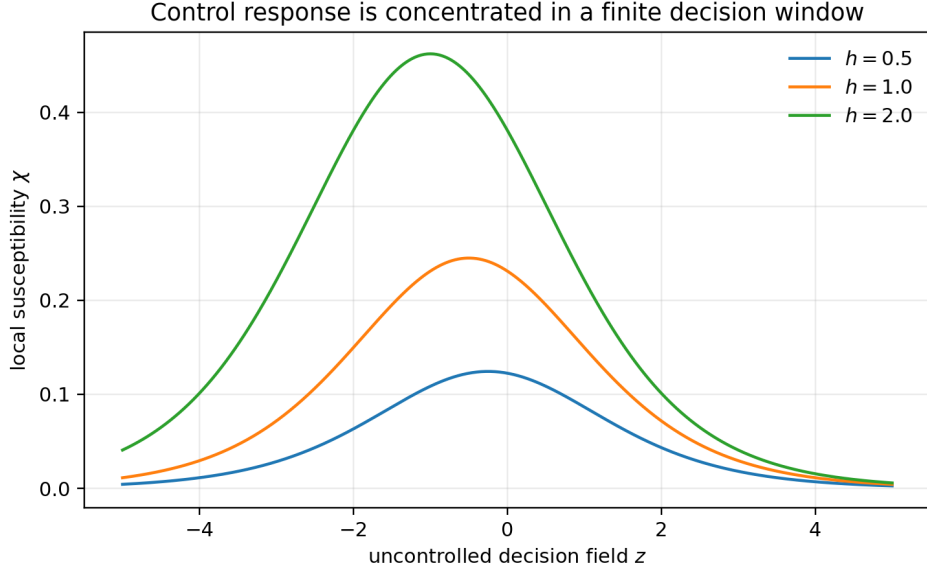


Figure 10: Exact local susceptibility for the logistic controlled transition. Strong response occurs only in a finite window of the uncontrolled decision field. This is why large actuation or strong epistemic bias can reduce useful response by pushing the system into saturation.

be the probability that a population with social state x is in the high-information regime. The epistemic-state-averaged susceptibility at that social state is

$$\bar{\chi}_x \equiv \mathbb{E}[\chi_{xK} \mid X_t = x] = (1 - r_x)\chi_{x0} + r_x\chi_{x1}. \quad (116)$$

Now use the elementary variance identity

$$\mathbb{E}[\chi_{xK}^2 \mid X_t = x] = \bar{\chi}_x^2 + \text{Var}(\chi_{xK} \mid X_t = x). \quad (117)$$

Because K is binary,

$$\text{Var}(\chi_{xK} \mid X_t = x) = r_x(1 - r_x)(\chi_{x1} - \chi_{x0})^2. \quad (118)$$

This simple formula is already informative: epistemic modulation vanishes if either epistemic state never occurs ($r_x = 0$ or 1) or if the controller affects both epistemic states equally ($\chi_{x1} = \chi_{x0}$).

9.3 Step 3: decompose the Pinsker lower bound

Equation (107) gives, at each full state $s = (x, k)$,

$$I(U; X' \mid X = x, K = k) \geq 2a_x(1 - a_x)\chi_{xk}^2. \quad (119)$$

Average this inequality first over K conditional on $X = x$:

$$\sum_k \mathbb{P}(k \mid x) I(U; X' \mid x, k) \geq 2a_x(1 - a_x) \sum_k \mathbb{P}(k \mid x) \chi_{xk}^2. \quad (120)$$

Now substitute Eq. (117):

$$\sum_k \mathbb{P}(k \mid x) I(U; X' \mid x, k) \geq 2a_x(1 - a_x) [\bar{\chi}_x^2 + \text{Var}(\chi_{xK} \mid x)]. \quad (121)$$

Finally average over the current social state X_t to obtain

$$I(U_t; X_{t+1} \mid X_t, K_t) \geq 2 \sum_x \mathbb{P}(X_t = x) a_x(1 - a_x) [\bar{\chi}_x^2 + \text{Var}(\chi_{xK} \mid X_t = x)]. \quad (122)$$

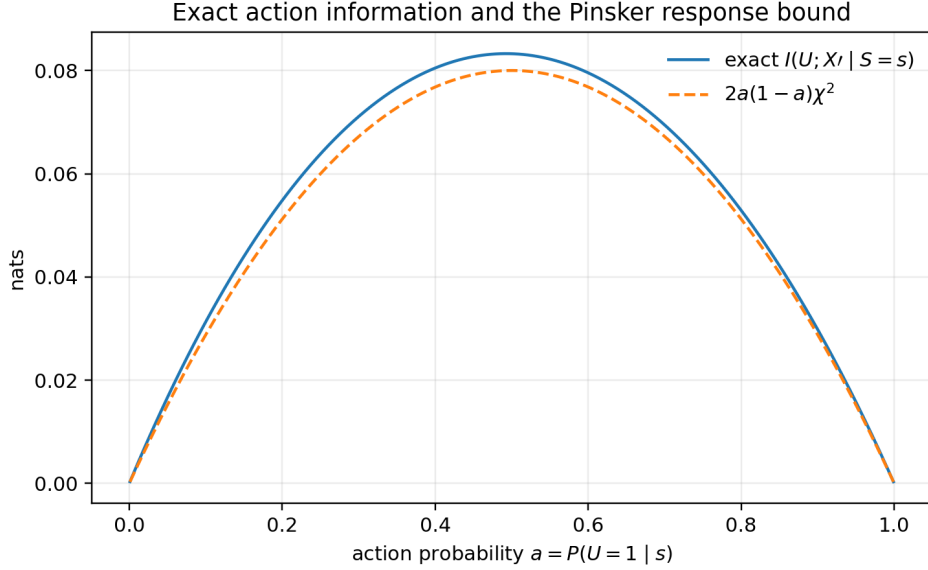


Figure 11: Example of Eq. (107) with $p_0 = 0.25$ and $p_1 = 0.65$. The exact conditional mutual information lies above the Pinsker response term for every action probability. Both vanish when the policy almost never or almost always takes one action, illustrating the importance of action support.

Using Eq. (118), this becomes

$$I(U_t; X_{t+1} | X_t, K_t) \geq 2 \sum_x \mathbb{P}(X_t = x) a_x (1 - a_x) [\bar{\chi}_x^2 + r_x (1 - r_x) (\chi_{x1} - \chi_{x0})^2]. \quad (123)$$

This is the missing bound we wanted: the conditional controller information must be large enough not only to support an average control response $\bar{\chi}_x$, but also to support the *epistemic dependence of that response*.

The two pieces have different meanings:

$$\text{mean-response piece} = 2 \sum_x \mathbb{P}(x) a_x (1 - a_x) \bar{\chi}_x^2, \quad (124)$$

$$\text{epistemic-modulation piece} = 2 \sum_x \mathbb{P}(x) a_x (1 - a_x) \text{Var}(\chi_{xK} | x). \quad (125)$$

The second term is the explicit measure of how strongly controllability varies across epistemic states.

9.4 Step 4: an epistemic modulation efficiency

Equation (122) suggests a bounded normalized quantity,

$$\eta_{\text{epi}}(t) \equiv \frac{2 \sum_x \mathbb{P}(X_t = x) a_x (1 - a_x) \text{Var}(\chi_{xK} | X_t = x)}{I(U_t; X_{t+1} | X_t, K_t)} \leq 1. \quad (126)$$

We do not give Eq. (126) a thermodynamic interpretation at this stage. Operationally it answers a clean question:

How much of the conditional controller information can be attributed specifically to the fact that the same action has different effects in different epistemic regimes?

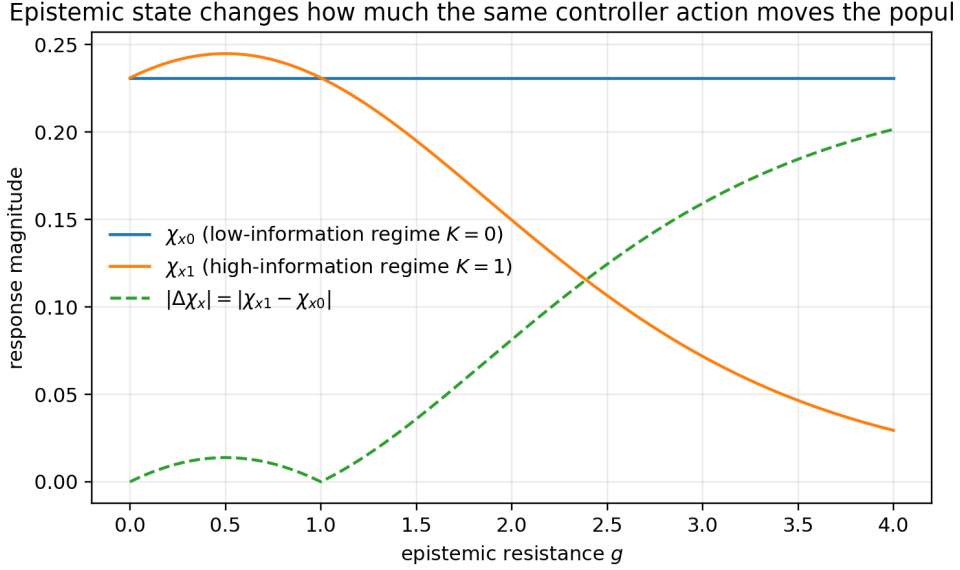


Figure 12: Epistemic modulation of local susceptibility for a fixed social state $x = 0$. The low-information and high-information populations respond differently to the same intervention. The dashed curve shows the modulation gap $|\Delta\chi_x| = |\chi_{x1} - \chi_{x0}|$.

This makes precise the informal idea of *epistemic resistance*: if high-information populations are harder to move than low-information populations, then the variability of χ_{xK} across K consumes part of the conditional action information budget.

9.5 Remark: state before or after the day?

In the present four-state model we conditioned on K_t , the epistemic state at the beginning of the round. In the full day/night dashboard system, information also spreads during the day before the controller acts. A more faithful later theory may therefore condition on a post-day epistemic variable, say $\tilde{K}_{t+1}$, and study

$$I(U_t; X_{t+1} \mid X_t, \tilde{K}_{t+1}). \quad (127)$$

The logic of the decomposition would be the same. We start with K_t here only because it keeps the finite-state model exact and transparent.

10 A second bound: the behavioral value of sensing

The previous bound starts from $I(U; X' \mid S)$ and asks how much action information appears as population response. A distinct question is: *how much advantage can feedback gain over an equally active open-loop controller simply because it knows the state?*

10.1 Matched open-loop reference

At a given round, define the effective feedback policy

$$\pi(u \mid s) = \mathbb{P}(U_t = u \mid S_t = s) \quad (128)$$

and its marginal action frequency

$$\bar{\pi}(u) = \sum_s \mu_t(s) \pi(u \mid s). \quad (129)$$

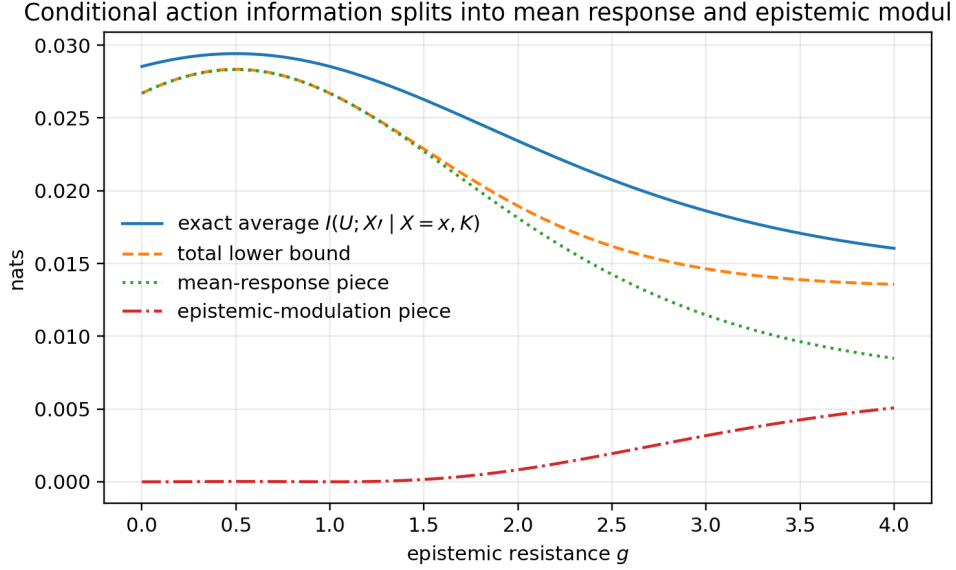


Figure 13: At a fixed social state x , the exact conditional action information (solid curve) exceeds the lower bound obtained by splitting the response into a mean-response contribution and an epistemic-modulation contribution. As epistemic resistance g increases, the controller affects low-information and high-information populations differently, and the modulation term grows.

Construct a matched open-loop controller that uses the same marginal action frequency but chooses U independently of S .

The feedback and matched-open-loop state-action distributions are

$$P(s, u) = \mu_t(s)\pi(u | s), \quad (130)$$

$$Q(s, u) = \mu_t(s)\bar{\pi}(u). \quad (131)$$

Their relative entropy is exactly

$$D_{\text{KL}}(P||Q) = \sum_{s,u} \mu_t(s)\pi(u | s) \ln \frac{\pi(u | s)}{\bar{\pi}(u)} \quad (132)$$

$$= \boxed{I(S_t; U_t)}. \quad (133)$$

Thus $I(S; U)$ is precisely the KL separation between state-dependent feedback actions and an equally active state-independent controller.

10.2 Propagate both controllers through the same dynamics

Let $W(s' | s, u)$ denote the next-state channel. Applying the same stochastic channel to P and Q cannot increase KL divergence, so

$$D_{\text{KL}}(P_{S'}||Q_{S'}) \leq I(S_t; U_t). \quad (134)$$

For any bounded observable $f(S') \in [0, 1]$, the difference in expectation is bounded by total variation, and Pinsker gives

$$|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq \text{TV}(P_{S'}, Q_{S'}) \leq \sqrt{\frac{1}{2}D_{\text{KL}}(P_{S'}||Q_{S'})}. \quad (135)$$

Taking $f(S') = X'$ and defining

$$\Delta_{\text{fb}} \equiv \mathbb{E}_{\text{feedback}}[X_{t+1}] - \mathbb{E}_{\text{matched open}}[X_{t+1}], \quad (136)$$

we obtain

$$\boxed{2\Delta_{\text{fb}}^2 \leq I(S_t; U_t) \leq I(S_t; Y_t)}. \quad (137)$$

The second inequality is Eq. (73).

Equation (137) is particularly useful conceptually. It says that the extra one-step behavioral effect obtained from *state-dependent* intervention, relative to an equally active open-loop controller, cannot exceed the information resource that the feedback loop actually uses, and that used information cannot exceed what was sensed.

Two bounded ratios follow naturally:

$$\eta_{\text{used}} = \frac{2\Delta_{\text{fb}}^2}{I(S; U)} \leq 1, \quad \eta_{\text{sens}} = \frac{2\Delta_{\text{fb}}^2}{I(S; Y)} \leq 1. \quad (138)$$

These are information-response efficiencies, not energetic efficiencies.

11 Exact numerical example

To make the formulas concrete, consider the parameter set in Table 1. The controller target is interpreted as false, so $g > 0$ represents epistemic resistance.

Table 1: Reference parameter set used in the figures below.

Parameter	Value	Meaning
ϵ	0.20	sensor error
π_0, π_1	0.85, 0.15	feedback policy
θ	-0.20	baseline social bias
J	1.00	social persistence
g	1.50	epistemic resistance
h	1.40	actuation strength
α	0.25	information acquisition
δ	0.12	information loss
μ_0	(0.45, 0.35, 0.10, 0.10)	initial four-state distribution

The effective action probabilities are

$$a_0 = 0.71, \quad a_1 = 0.29. \quad (139)$$

Initially $q_0 = P(X_0 = 1) = 0.20$ and $\kappa_0 = P(K_0 = 1) = 0.45$. Exact matrix propagation gives the trajectories in Fig. 14. The epistemic state relaxes to the analytically known value

$$\kappa^* = \frac{0.25}{0.25 + 0.12} \approx 0.676, \quad (140)$$

while the coupled social state approaches $q^* \approx 0.363$.

At $t = 0$ the exact information quantities are

$$I(S_0; Y_0) \approx 0.1265 \text{ nats}, \quad (141)$$

$$I(S_0; U_0) \approx 0.0589 \text{ nats}, \quad (142)$$

$$I(U_0; X_1 | S_0) \approx 0.0341 \text{ nats}. \quad (143)$$

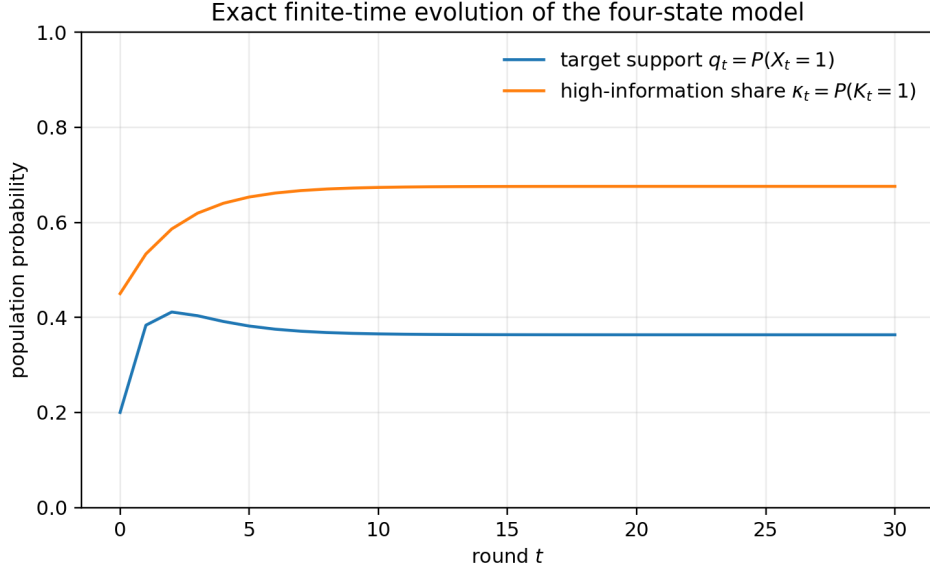


Figure 14: Exact finite-time population dynamics obtained from $\mu_{t+1} = \mu_t M$. No Monte Carlo sampling is used.

The Pinsker numerator in Eq. (109) is approximately 0.0286 nats, giving $\eta_{\text{IR}}(0) \approx 0.836$. By round 30, the corresponding values are approximately 0.1792, 0.0844, 0.0302, and 0.0232 nats, respectively, with $\eta_{\text{IR}} \approx 0.769$. Figure 15 shows the exact round dependence.

Finally, Fig. 16 varies only the epistemic resistance g and recomputes the exact stationary distribution. The result is the qualitative behavior desired from the model: a high-information population becomes progressively harder for a false controller to steer.

12 What has and has not been established

The four-state construction already provides several useful exact statements:

1. The sensing information is exactly

$$I(S_t; Y_t) = h(\epsilon + (1 - 2\epsilon)q_t) - h(\epsilon).$$

2. The controller cannot encode more state information in its actions than it obtained from sensing:

$$I(S_t; U_t) \leq I(S_t; Y_t).$$

3. The epistemically conditioned action-to-population information is available in closed form:

$$I(U_t; X_{t+1} \mid X_t, K_t) = \sum_{x,k} \mu_t(x, k) \{h(\bar{p}_{xk}) - (1 - a_x)h(p_{xk0}) - a_x h(p_{xk1})\}.$$

4. Omitting K_t necessarily loses or preserves, but cannot add, action-to-next-state information under this model:

$$I(U; X' \mid X, K) \geq I(U; X' \mid X).$$

5. Susceptibility and CMI are linked by a rigorous bounded efficiency:

$$2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2 \leq I(U_t; X_{t+1} \mid S_t).$$

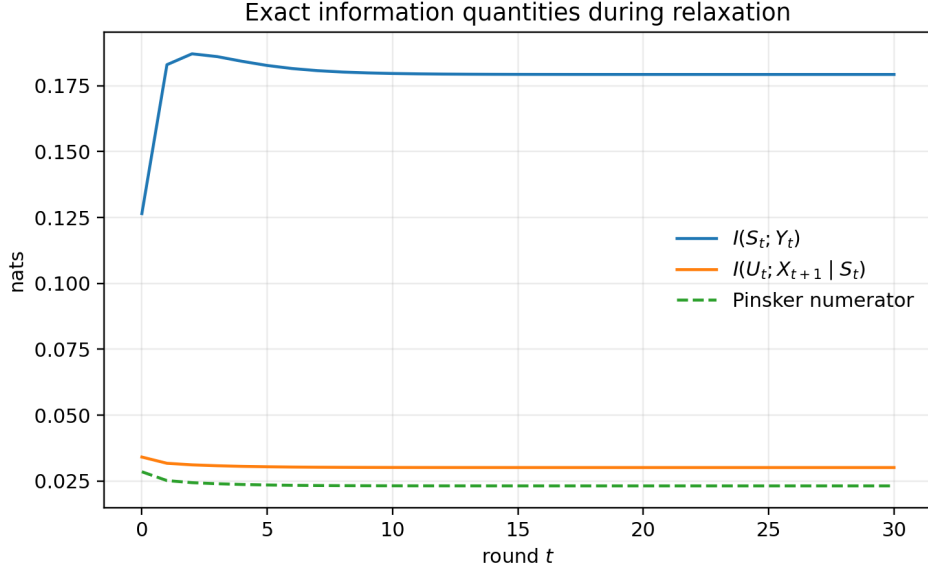


Figure 15: Exact sensing information, action-to-population CMI, and the Pinsker response numerator during relaxation of the reference model. The bound remains satisfied pointwise in time.

6. The behavioral advantage of feedback over a matched open-loop policy is bounded by sensing information:

$$2\Delta_{\text{fb}}^2 \leq I(S_t; U_t) \leq I(S_t; Y_t).$$

What we have *not* yet established is a stochastic-thermodynamic entropy balance of the form “system entropy change + currents + information = total entropy production.” Such a statement requires an explicit definition of reverse trajectories and, if one wants a thermodynamic interpretation rather than only a KL identity, a principled local-detailed-balance-like structure. This caution is consistent with the distinction emphasized in feedback thermodynamics between different information measures and their operational measurement implementations [1]. The social-imitation framework of Irisarri *et al.* provides a possible route for the later stage: define microscopically reversible social and epistemic transitions first, then derive path irreversibility and second-law-like inequalities from those transition ratios rather than postulating an energy function [2].

13 How this becomes the Stage-II mean-field theory

The next theory should replace the binary epistemic variable K by an epidemic-like distribution of explicit facts. Conceptually, the four-state objects already identify what must survive the generalization:

- X_t becomes a population vote fraction or magnetization m_t ;
- K_t becomes one or more coarse epistemic coordinates κ_t derived from fact occupancy;
- α and δ become fact-acquisition and forgetting terms generated by dashboard sampling and persistence;
- the exact finite-state kernel becomes a mean-field map plus finite- N noise;
- $I(S; Y)$, $I(S; U)$, $I(U; X' | S)$, χ , and the two bounds derived above remain the quantities to reproduce or approximate.

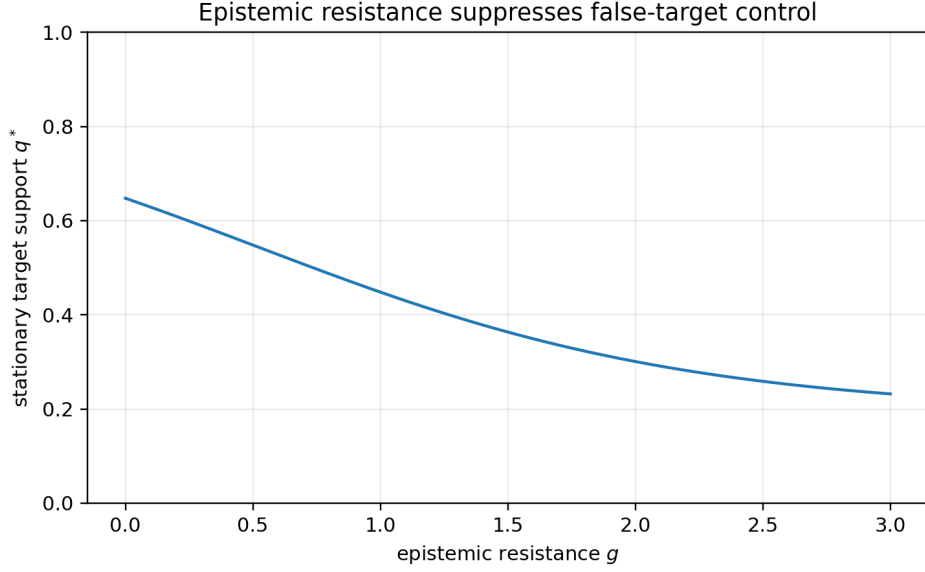


Figure 16: Exact stationary target support versus epistemic resistance. Increasing g shifts the controlled population away from the false controller target while leaving the autonomous K -process stationary fraction unchanged.

The benefit of the present dummy theory is that the meaning of each information quantity is fixed before the state space becomes large.

14 Summary

The four-state model is intentionally minimal, but it already separates three logically distinct stages of feedback control:

$$\begin{array}{lcl}
 \text{sensed information: } I(S; Y) & \longrightarrow & \text{state-dependent action: } I(S; U) \\
 & \longrightarrow & \text{population response: } I(U; X' | S), \chi
 \end{array} \tag{144}$$

The first data-processing inequality limits how much sensed information can survive in the policy. The second family of inequalities limits how much of the action-dependent information can appear as a coherent population shift. These results provide a clean exact reference for later mean-field and stochastic-thermodynamic extensions with explicit epistemic fact dynamics.

References

- [1] J. M. Horowitz and H. Sandberg, “Second-law-like inequalities with information and their interpretations,” arXiv:1409.5351 (2014).
- [2] L. Irisarri, L. Trigal, R. Toral, and G. Manzano, “Stochastic Thermodynamics of Social Imitation beyond Energetics,” arXiv:2511.14006v2 (2026).